

MAKSATIKHA, Russian Federation

WMO: 272080

Lat: 57.800N

Lon: 35.883E

Elev: 466

StdP: 14.45

Time Zone: 3.00 (E03)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.1	-8.4	-18.7	2.0	-13.7	-12.8	2.8	-8.0	14.6	25.0	13.6	25.3	1.9	60	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	84.6	68.0	80.5	66.6	76.9	64.4	70.0	81.5	68.1	77.5	66.2	74.4	5.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	97.3	74.3	64.4	92.1	72.9	62.9	87.2	70.8	34.2	82.0	32.6	77.4	31.2	74.5	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.3	11.8	10.5	DB	-22.7	88.1	7.8	4.1	-28.3	91.1	-32.8	93.5	-37.2	95.8	-42.8	98.8	(4)
(5)			WB	-22.9	71.9	7.7	2.1	-28.5	73.4	-33.0	74.6	-37.3	75.8	-42.9	77.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.1	17.1	18.2	27.5	40.4	54.2	59.5	65.2	61.1	51.7	40.8	31.3	24.9		
	DBStd	18.71	13.37	12.69	9.42	7.47	8.13	6.47	5.96	6.63	6.07	7.77	8.53	10.79		
	HDD50	4649	1021	891	697	300	47	5	0	3	51	296	560	778		
	HDD65	8857	1486	1311	1162	738	345	189	68	156	399	750	1010	1243		
	CDD50	1415	0	0	0	12	178	290	473	349	102	11	0	0		
	CDD65	150	0	0	0	0	12	24	76	37	1	0	0	0		
	CDH74	1471	0	0	0	0	148	217	723	376	7	0	0	0		
	CDH80	400	0	0	0	0	31	42	215	112	0	0	0	0		
	WSAvg	5.1	5.5	5.3	5.6	5.0	5.0	4.7	4.1	4.1	4.6	5.1	5.7	6.0		
	PrecAvg	25.3	1.8	1.5	1.2	1.4	2.2	2.9	2.9	2.7	2.1	2.5	2.2	2.0		
(15) Precipitation	PrecMax	32.3	3.1	2.7	1.8	2.7	5.0	6.6	6.8	6.6	4.8	4.3	4.8	3.3		
	PrecMin	18.4	0.5	0.3	0.4	0.4	0.6	0.1	0.2	0.0	0.4	0.6	0.1	1.0		
	PrecStd	3.9	0.7	0.6	0.4	0.6	1.1	1.3	1.9	1.3	1.1	1.1	1.1	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.3	38.3	54.4	70.0	84.3	85.3	91.5	90.1	74.0	63.4	50.1	46.0		
		MCWB	38.7	35.1	43.0	51.0	66.6	66.9	70.6	72.0	62.1	54.5	47.9	44.7		
	2%	DB	36.8	36.0	47.3	63.9	77.9	80.1	85.7	82.5	69.3	56.2	46.0	42.5		
		MCWB	35.4	33.5	39.2	49.1	62.1	65.6	69.1	67.3	60.5	51.7	43.6	41.0		
	5%	DB	34.8	34.5	42.3	58.5	73.4	75.7	81.4	77.8	65.7	53.9	43.7	39.2		
		MCWB	33.3	32.7	37.1	46.8	59.6	63.4	67.3	66.3	57.2	50.7	41.9	37.5		
	10%	DB	33.2	33.2	39.1	54.1	69.0	72.1	77.4	73.6	62.5	51.6	41.6	36.2		
		MCWB	32.0	31.8	35.3	44.8	56.3	61.3	64.9	63.7	56.2	49.1	40.0	34.7		
	0.4%	WB	39.2	36.0	44.7	53.1	69.2	70.1	71.9	72.8	65.0	55.8	48.6	44.8		
		MCDB	40.1	37.4	54.5	64.3	82.7	80.1	86.2	83.7	70.8	59.3	49.3	45.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.5	34.2	40.3	50.8	64.3	67.6	70.1	69.8	61.0	53.0	44.6	41.1		
		MCDB	36.4	35.2	44.8	60.8	74.3	77.8	82.6	80.2	67.2	55.1	45.7	42.3		
	5%	WB	33.6	33.1	37.8	48.3	61.0	65.1	68.3	67.4	58.9	51.1	42.2	37.7		
		MCDB	34.3	34.1	41.5	57.4	69.8	72.5	78.2	75.3	64.6	53.6	43.4	39.0		
	10%	WB	32.0	31.9	35.7	45.5	58.5	62.8	66.7	65.0	56.7	49.1	40.1	35.0		
		MCDB	33.3	33.1	38.9	53.3	66.8	69.8	75.3	72.0	61.8	51.6	41.4	36.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	11.5	13.8	17.5	19.3	17.8	18.4	18.0	15.2	10.5	6.6	7.3		
		MCDBR	6.2	7.7	17.4	26.5	26.2	22.7	24.3	25.1	21.6	15.3	8.8	7.5		
		MCWBR	6.2	6.6	11.3	14.2	12.0	10.6	9.9	11.8	12.6	10.6	7.9	7.3		
		MCDBR	5.8	7.1	15.2	23.2	22.8	19.6	20.8	21.8	18.4	12.7	7.7	7.1		
	5% WB	MCWBR	6.0	6.5	10.5	13.2	11.4	10.4	9.6	11.1	11.5	9.4	7.3	7.0		
(40) Clear-Sky Solar Irradiance	taub		0.279	0.318	0.336	0.386	0.377	0.366	0.401	0.399	0.362	0.340	0.311	0.270		
	taud		2.104	2.075	2.139	2.211	2.363	2.412	2.330	2.324	2.403	2.379	2.224	2.098		
	Ebn at Noon		174	219	253	259	272	277	263	254	246	217	163	140		
	Edh at Noon		17	28	35	38	35	34	36	34	27	21	16	13		
(44) All-Sky Solar Radiation	RadAvg		140	388	863	1260	1644	1703	1664	1312	810	354	123	74		
	RadStd		23	68	105	137	134	150	210	114	110	68	20	18		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page